Applied Reasoning Task

Identifying Machine Learning Problems

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1. Introduction

Before developing a machine learning model, it is important to correctly understand the problem that needs to be solved. Machine learning problems can differ significantly depending on whether the available data contains known outcomes, what type of prediction is required, and whether predefined groups exist.

This report analyzes three business scenarios and identifies whether each problem is supervised or unsupervised learning, whether it is a classification or regression problem where applicable, and what the likely features and target variables would be. An additional real-world scenario is also proposed. Finally, one important risk of incorrectly applying machine learning is discussed.

2. Scenario 1: Predicting Customer Churn

Business Scenario

A telecommunications company wants to identify customers who are likely to cancel their subscription. The company can use this information to contact high-risk customers and offer suitable services or incentives to retain them.

Learning Type

Supervised Learning

This is a supervised learning problem because the company can use historical customer data where the outcome is already known. For previous customers, the company knows whether they eventually cancelled their subscription or continued using the service.

Problem Type

Classification

The problem is classification because the model predicts a category rather than a continuous numerical value.

The possible outcomes are:

- **Yes** – Customer will churn.
- **No** – Customer will not churn.

Likely Features

Possible features include:

- Customer age
- Monthly bill amount
- Contract type
- Length of time as a customer
- Number of complaints
- Internet or data usage
- Number of customer service interactions
- Payment history

Target Variable

Target: Whether the customer will churn (**Yes/No**).

Reasoning

This is supervised learning because historical examples contain known customer outcomes. It is classification because the model predicts one of two categories: churn or no churn.

3. Scenario 2: Predicting House Prices

Business Scenario

A real-estate company wants to estimate the selling price of a house based on its characteristics and location. The prediction can help the company and customers make better pricing decisions.

Learning Type

Supervised Learning

This is supervised learning because historical house records contain known selling prices. These previous examples can be used to learn the relationship between house characteristics and their prices.

Problem Type

Regression

This is a regression problem because the target is a continuous numerical value. The model may predict values such as Rs. 50 lakh, Rs. 75 lakh, or Rs. 1.2 crore.

Likely Features

Possible features include:

- Location
- Area in square feet
- Number of bedrooms
- Number of bathrooms
- Age of the property
- Number of floors
- Parking availability
- Nearby schools and hospitals
- Distance from public transportation

Target Variable

Target: Selling price of the house.

Reasoning

This is supervised learning because the model is trained using historical houses for which the actual selling price is known. It is regression because the output is a numerical value that can vary over a continuous range.

4. Scenario 3: Customer Segmentation

Business Scenario

An e-commerce company wants to divide its customers into different groups based on their purchasing behavior. The company can then design different marketing strategies for each group.

Learning Type

Unsupervised Learning

This is an unsupervised learning problem because the company does not have predefined labels for its customers. Instead, the algorithm must discover patterns and groups within the available customer data.

Problem Type

Clustering

The problem is clustering because the objective is to automatically group customers who have similar characteristics or purchasing behavior.

For example, the algorithm might discover groups such as:

- Frequent high-value customers
- Occasional customers
- New customers
- Customers with low purchase activity

These groups are discovered from the data rather than being predefined during training.

Likely Features

Possible features include:

- Purchase frequency
- Average order value
- Total spending
- Number of products purchased
- Recency of last purchase
- Product categories purchased
- Number of orders

Target Variable

Target: There is no predefined target variable.

The algorithm identifies groups based on similarities among the input features.

Reasoning

This is unsupervised learning because the dataset does not contain a known target label. The purpose is to discover hidden patterns or natural groups within the customer population. Clustering algorithms can therefore be used to create meaningful customer segments.

5. Scenario 4: Predicting Hospital Patient Readmission

Business Scenario

A hospital wants to predict whether a patient is likely to be readmitted within 30 days after being discharged. The objective is to identify higher-risk patients so that the hospital can provide appropriate follow-up care and monitoring.

Learning Type

Supervised Learning

This is supervised learning because the hospital can use historical patient records where the outcome is already known. For previous patients, the hospital knows whether they were readmitted within 30 days after discharge.

Problem Type

Classification

This is a classification problem because the model predicts one of two categories:

- **Yes** – Patient is readmitted within 30 days.
- **No** – Patient is not readmitted within 30 days.

Likely Features

Possible features include:

- Patient age
- Previous hospital admissions
- Length of previous hospital stay
- Number of emergency visits
- Number of medications prescribed
- Previous diagnoses
- Number of follow-up visits
- Recent laboratory test results
- Discharge condition
- Time since the previous admission

Target Variable

Target: Whether the patient is readmitted within 30 days (**Yes/No**).

Reasoning

This is supervised learning because historical patient records contain known outcomes regarding readmission. It is classification because the model predicts a categorical outcome: whether the patient will be readmitted or not.

The prediction could help hospitals identify patients who may require additional follow-up or monitoring. However, such predictions should support healthcare professionals rather than replace clinical judgment.

6. Summary of the Four Scenarios

The four scenarios can be summarized as follows:

Scenario	Learning Type	Problem Type	Target
Customer Churn	Supervised	Classification	Churn: Yes/No
House Price Prediction	Supervised	Regression	House selling price
Customer Segmentation	Unsupervised	Clustering	No predefined target
Hospital Patient Readmission	Supervised	Classification	Readmission: Yes/No

7. Risk of Misapplying Machine Learning

One important risk of misapplying machine learning is making incorrect decisions when a model's predictions are treated as completely reliable.

For example, consider the hospital patient readmission scenario. If the model incorrectly predicts that a high-risk patient will not be readmitted, healthcare professionals may fail to provide additional follow-up care that could have been beneficial. On the other hand, if the model incorrectly identifies a low-risk patient as high-risk, hospital resources may be unnecessarily allocated to that patient.

There can also be problems if the training data is biased or incomplete. For example, if historical patient records do not adequately represent different patient groups, the model may perform well for some groups but poorly for others. This could result in unfair or unsafe decisions.

Therefore, machine learning predictions should be treated as decision support rather than absolute truth, especially in high-impact areas such as healthcare. The quality of the training data, model performance, and real-world consequences of incorrect predictions should be carefully considered before deploying the system.

8. Conclusion

Correctly identifying the type of machine learning problem is an essential step before selecting an algorithm or writing any code. The business objective should determine whether supervised or unsupervised learning is appropriate and whether the problem requires classification, regression, or clustering.

In the scenarios discussed, customer churn and hospital readmission are classification problems, house price prediction is a regression problem, and customer segmentation is an unsupervised clustering problem.

The hospital readmission scenario also demonstrates why careful reasoning is important before applying machine learning. A technically accurate model can still cause problems if the data is biased, the problem is incorrectly formulated, or the predictions are used without considering their real-world impact.

The main lesson is that **the business problem should be understood before choosing**

the machine learning approach. This reasoning forms the foundation for responsible and effective machine learning.